

Learning Robust Representation in Tonic-Clonic Seizures with Cyclic Transformer

Jingwei Zhang, Lauren Swinnen, Christos Chatzichristos, Wim Van Paesschen, and Maarten De Vos

Abstract—Tonic-clonic seizures (TCSs) pose a significant risk for sudden unexpected death in epilepsy (SUDEP). Previous research has highlighted the potential of multimodal wearable seizure detection systems in accurately detecting TCSs through continuous monitoring, enabling timely alarms and potentially preventing SUDEP. However, such multimodal systems carry a higher risk of sensor malfunction. In this paper, we propose a cyclic transformer approach to address these challenges. The cyclic transformer learns a robust representation by performing circular modal translations between the source and target modalities. It leverages back-translation as regularization technique to enhance the discriminative power of the learned representation. Notably, the proposed cyclic transformer is trained on paired multimodal data but requires only a single source modality during deployment. This characteristic ensures the robustness of the cyclic transformer to perturbations or missing information in the target modality. Experimental results demonstrate that the proposed cyclic transformer achieves competitive performance compared with existing multimodal systems. While both approaches were trained using EEG and EMG data, the cyclic transformer exclusively employs EEG data for testing, diverging from the state-of-the-art's utilization of both EEG and EMG data during test. This showcases the effectiveness of the cyclic transformer in multimodal TCSs detection, offering a promising approach for enhancing the accuracy and robustness of seizure detection systems while mitigating the risks associated with sensor malfunction.

Index Terms—Automated seizure detection, co-learning, epilepsy, tonic clonic seizures.

I. INTRODUCTION

Tonic-clonic seizures (TCSs), characterized by stiffening of the body followed by jerking of the limbs, are widely recognized as one of the most severe forms of seizures, with potentially life-threatening consequences [1]. The gold standard for TCSs detection is video-electroencephalography (video-EEG) [2]. However, the adoption of video-EEG for long-term monitoring is limited due to the cost and the discomfort associated with wearing the full scalp equipment [3, 4].

Patients with TCSs would benefit from long-term monitoring with wearables which can give an alarm when a seizure occurs [5-18]. Especially the multimodal wearable systems, have gained significant attention [11-14, 16, 17].

Multimodal TCSs detection systems integrate multiple modalities that are relevant to the different semiological phases of TCSs and leverage their individual strengths and complementarity, resulting in enhanced performance. For example, combining electrodermal activity (EDA) and electrocardiogram (ECG) modalities enables to detect the TCS by identifying both changes in sympathetically mediated EDA and the heart rate variability associated with TCSs [11]; combining electromyography (EMG) and accelerometer (ACC) may enhance the detection performance by leveraging both the muscular information and movement information [13].

However, this implementation of multimodal systems usually necessitates multiple devices. The requirement of multiple sensors not only increases the overall costs associated with these systems but also introduces the risk of missing modalities [19]. Notably, modalities such as EMG and EDA, which involve electrodes positioned on muscles, are susceptible to disconnection during physical activities, heightening the possibilities of missing data [20]. As the potential of missing modalities significantly impacts the functioning of multimodal systems and is difficult to completely avoid, the widespread adoption of multimodal systems becomes challenging.

The possibility of missing modalities requires multimodal seizure detection algorithms to disengage from the common assumption that all modalities should be available, aligned, and noiseless [19]. Previous research addressed this challenge by reconstructing the data via data imputation [21-23]. Some studies have successfully reconstructed the missing data via matrix completion methods [24, 25], generative models [26-29], and meta learning [30].

However, data imputation on biomedical signals, especially those often used in seizure detection, is particularly challenging due to the high variability exhibited by biomedical

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SeizelT2 project was approved by the ethical committee of UZ/KU Leuven (internal number: S63631). Written informed consent was obtained for every participant.

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signals like EEG and EMG [31].

In this paper, we propose a novel approach called cyclic transformer that learns intermediate representation of two modalities. Departing from conventional methodologies focused on modalities reconstruction, the proposed cyclic transformer employs a circular modality translation to learn an intermediate representation between the source and target modalities. The cyclic transformer introduces a novel application of transformer architecture between the context of learning intermediate representation from two distinct biomedical modalities. In comparison to the vanilla transformer, the cyclic transformer has an additional classifier and reconstructor, incorporating regularization mechanisms. Leveraging this regularization technique, the approach provides more robust and discriminative representations through circular modality translation, making it more accurate in seizure detection.

A notable feature of the cyclic transformer is its robustness in the presence of incomplete data, a frequent challenge encountered in long-term epilepsy monitoring. This robustness stems from its distinctive characteristic: during deployment, the target modalities are not required to obtain the representation. Consequently, the cyclic transformer presents a notable advantage in handling incomplete datasets within the realm of biomedical signal processing. It adeptly circumvents the challenges in reconstructing missing modalities. This pivotal attribute significantly enhances its effectiveness, especially when confronted with biomedical signals that inherently exhibit substantial variability.

The rest of this paper is organized as follows. Section II provides an overview of state-of-the-art approaches in TCS detection. In section III we present the proposed cyclic transformer and its multiple input version. Then we conduct several experiments to evaluate the effectiveness of the proposed cyclic transformer. The dataset used for evaluation, benchmark models, variations of the proposed cyclic transformer, and the evaluation metrics are introduced in section IV. Then, the results are presented and discussed in section V.

II. RELATED WORK

In recent years, the state-of-the-art in TCS detection has witnessed significant achievements. Previous studies explored various modalities on its own or combined, such as EEG, EMG, ACC, ECG, and EDA for TCS detection, demonstrating their potential.

Signal processing techniques, such as time-frequency analysis and wavelet transforms, played a crucial role in extracting meaningful features associated with TCS. Notable achievements include the study by Conradsen et al., which achieved a sensitivity of 100.0% and a false positive rate of 1.0/24 hours through counting the number of zero crossings in EMG signals [32].

Machine learning classifiers, such as support vector machines and logistic regression, have been employed to capture TCS patterns when there is a more extensive set of features. For example, Kusmakar et al. achieved a state-of-the-

art performance by employing a support vector data description classifier on 21 features extracted from ACC signals. Their approach demonstrated a sensitivity of 95.2% and a false positive rate of 0.72/24 hours [8].

Integrating multiple modalities has proven advantageous in improving TCS detection performance. Milosevic et al. demonstrated notable performance by achieving a sensitivity of 91.0% and a false positive rate of 1.0/24 hours through the integration of features extracted from EMG and ACC [13]. Onorati et al. further highlighted the potential of combining EDA with ACC, showcasing an algorithm with a sensitivity of 94.5% and a false positive rate of 0.19/24 hours [33]. In a similar context, we further explored the incorporation of wearable EEG signals. Our approach showed a remarkable sensitivity of 97.7% and a false positive rate of 0.4/24 hours. Expanding on this integration approach, we combined ACC with wearable EEG and EMG, achieving a sensitivity of 90.9% and an impressively low false positive rate of 0.1/24 hours [17].

While these studies were conducted on different datasets, which makes it challenging for direct comparison, some insights can be learned. For example, the ability of processing a larger number of complex features would be advantageous for TCS detection approaches. Consequently, employing machine learning methods, and potentially deep learning, may enhance TCS detection approaches. Additionally, the integration of multiple modalities has demonstrated advantages, suggesting that further development of more effective multimodal approaches holds promise for improving TCS detection performance.

III. METHOD

A. Transformer backbone

A transformer is a sequence transduction model that is based solely on attention mechanisms [34]. As shown in Fig. 1 (a), a transformer has an encoder-decoder structure, where the encoder first maps an input sequence into an embedding and then the decoder projects the embedding into an output sequence.

The transformer encoder is formed by stacking multiple transformer encoder layers. Each transformer encoder layer consists of two sublayers: a multi-head self-attention neural network and a position-wise feed-forward neural network. Similarly, the transformer decoder is also constructed as a stack of transformer decoder layers. A transformer decoder layer includes the two sub-layers found in the encoder layer and an extra sub-layer, which conducts multi-head attention over the embedding and the masked output sequence. For both encoder and decoder layer, residual connections and layer normalization are employed within each sublayer to stabilize and accelerate training.

The multi-head attention neural network is used to re-weight each token based on its relevance to other tokens in the sequence and can be described as

$$att_i(X) = softmax\left(\frac{W_i^K X (W_i^Q X)^T}{\sqrt{d_k}}\right) W_i^V \quad (1)$$

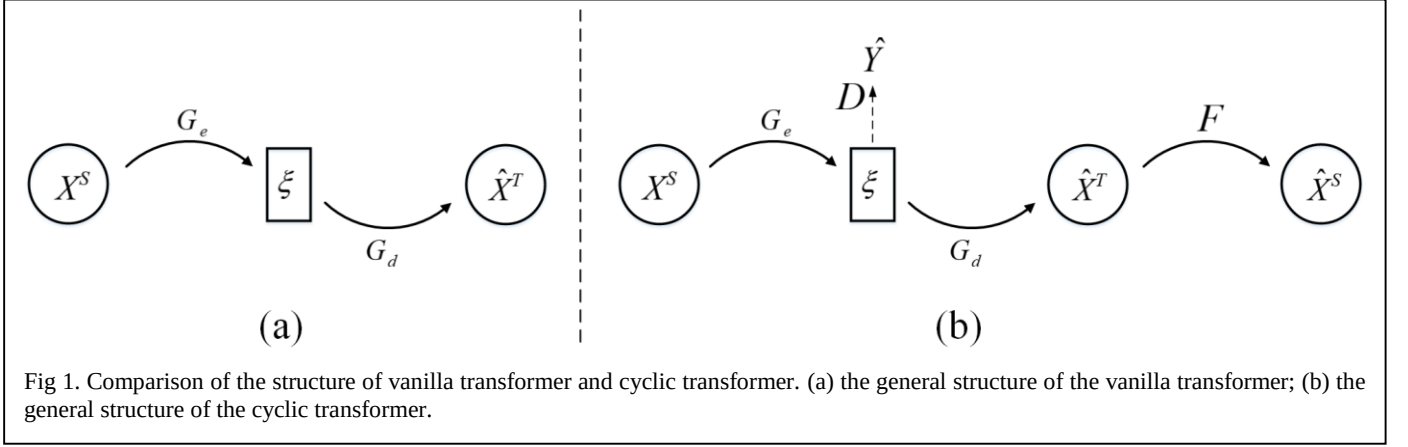


Fig 1. Comparison of the structure of vanilla transformer and cyclic transformer. (a) the general structure of the vanilla transformer; (b) the general structure of the cyclic transformer.

$$\text{multihead}(X) = \text{concat}(\text{att}_1, \dots, \text{att}_H)W^O \quad (2)$$

$$Z = \text{layernorm}(X + \text{multihead}(X)) \quad (3)$$

where the input $X \in R^{d_{\text{batch}} \times d_{\text{seq}} \times d_{\text{feat}}}$ is a batch of d_{seq} tokens with d_{feat} features in each token. The weights have the dimension $W^K, W^Q, W^V \in R^{d_{\text{feat}} \times d_k}$, $W^O \in R^{d_k \times H \times d_{\text{feat}}}$, where d_k equal to d_{feat} divide the number of heads H .

The position-wise feed-forward neural network denotes as

$$\text{ffn}(Z) = \max(0, ZW_1^F + b_1^F)W_2^F + b_2^F \quad (4)$$

$$\text{output} = \text{layernorm}(Z + \text{ffn}(Z)) \quad (5)$$

where the weights $W_1^F \in R^{d_{\text{feat}} \times d_{\text{ffn}}}$, $W_2^F \in R^{d_{\text{ffn}} \times d_{\text{feat}}}$ and the biases $b_1^F \in R^{d_{\text{ffn}}}$, $b_2^F \in R^{d_{\text{feat}}}$.

B. Cyclic Transformer

Given a bi-modal training set $X = \{\{X_1^S, X_1^T\}, \dots, \{X_L^S, X_L^T\}\}$, where L denotes the number of samples, X^S is the source modality and X^T is the target modality, either X_i^S or X_i^T consists of d_{seq} tokens, the objective of cyclic transformer is to identify the label $Y \in \{0, 1\}$ of each token in a sample.

The proposed cyclic transformer, as shown in Fig. 1 (b), has an encoder-decoder-reconstructor architecture. It uses the transformer as backbone but could do the classification task as a classifier is integrated into its architecture.

i. Task-aware reconstruction

We use a transformer, which has two transformer encoder layers and two transformer decoder layers, as a backbone model to conduct a sequence transduction task between source modality and target modality. By this means, an intermediate representation that captures joint information between different modalities can be learned. However, the vanilla transformer backbone cannot directly be used for classification tasks. To adapt the transformer for classification tasks, the original transformer backbone is modified in the following ways:

- The additional sub-layer in the decoder layer is removed, making the encoder layer and the decoder layer have the same architecture.
- A classifier is integrated into the architecture to provide

predictions based on the intermediate representation.

Given a batch of input sequences $X^S \in R^{d_{\text{batch}} \times d_{\text{seq}} \times d_{\text{feat}}}$, a positional encoding is added to it to inject the positional information of the tokens in the sequences as the transformer layer is insensitive to the ordering of input. The positional encodings have the same dimension as the embedded inputs and therefore can be directly summed to it.

$$\tilde{X}^S = X^S + P \quad (6)$$

There are many techniques for positional encoding [35-37]. We chose a fixed encoding that use sine and cosine functions of different frequencies to indicate the positional information. For each sequence in the batch,

$$P(\text{pos}, 2i) = \sin\left(\frac{\text{pos}}{10000^{\frac{2i}{d_{\text{feat}}}}}\right) \quad (7)$$

$$P(\text{pos}, 2i + 1) = \cos\left(\frac{\text{pos}}{10000^{\frac{2i}{d_{\text{feat}}}}}\right) \quad (8)$$

where $1 \leq \text{pos} \leq d_{\text{seq}}$ denotes the position, $1 \leq i \leq d_{\text{feat}}$ denotes the dimension.

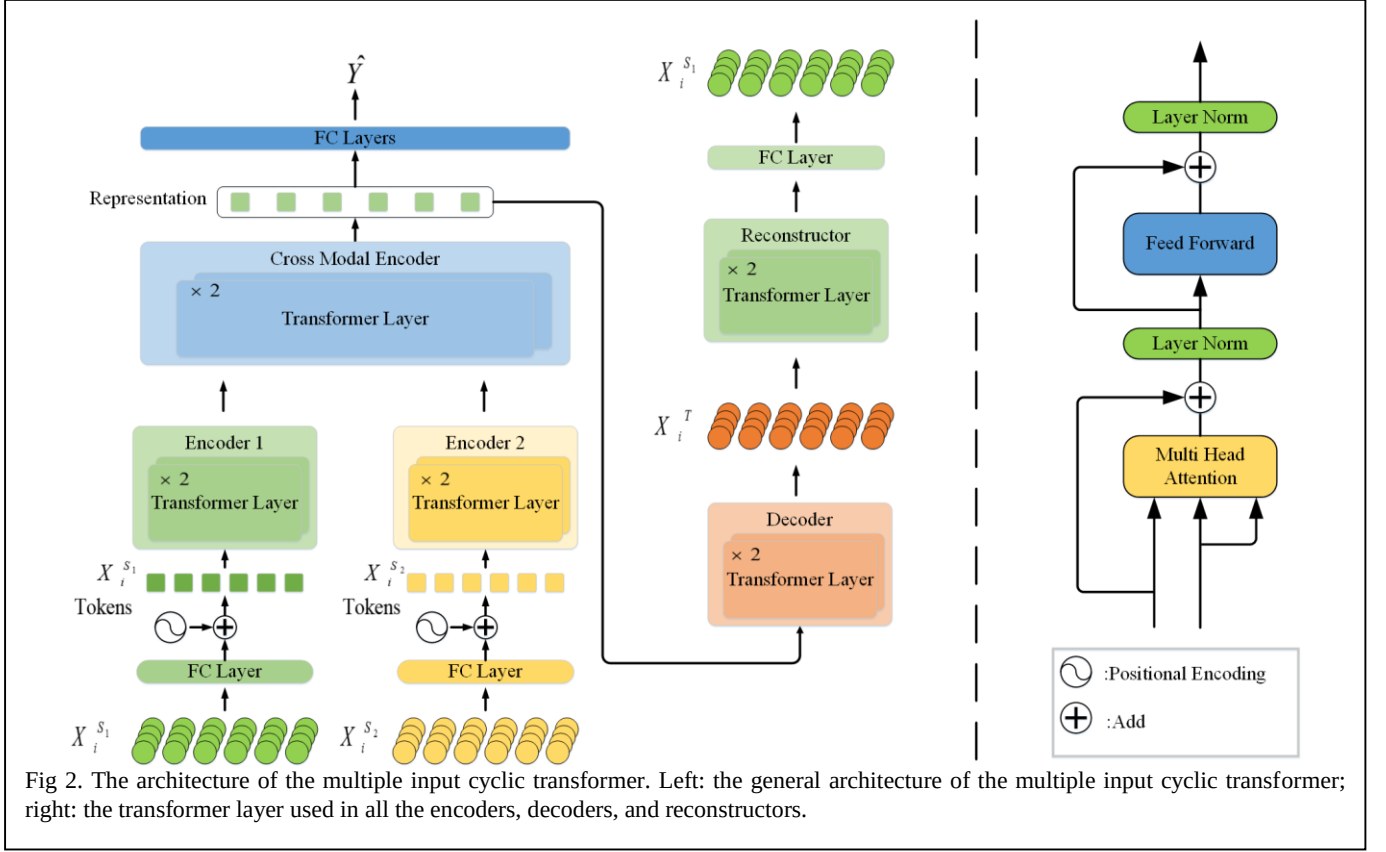
Once the positional encoding is done, the transformer encoder G_e maps it into an embedding ξ

$$\xi = G_e(\tilde{X}^S) \quad (9)$$

and then the decoder G_d projects the embedding into the target modality sequence $\hat{X}^T$

$$\hat{X}^T = G_d(\xi) \quad (10)$$

Like other encoder-decoder architectures, the intermediate representation learned by the transformer might include redundant information that does not directly contribute to the final classification [38]. The task-aware transformer introduces an additional regularization by integrating a classifier into its architecture. This regularization motivates the joint representation to exclusively preserve the relevant information required for the classification task, leading to improved efficiency and effectiveness in the classification process.



The classifier D takes the intermediate representation ξ as input and output the prediction y :

$$Y = D(\xi) \quad (11)$$

where D is a stack of two fully connected layers in this paper.

ii. Cyclic transformer: learning with back-translation

Although intermediate representation can be learned through the translation from X^S to X^T , the network may struggle to capture and preserve the shared characteristics due to the lack of robustness. As a result, the learned intermediate representation may be less reliable and more sensitive to noise.

Inspired by the cycle GAN [39] and the Multimodal Cyclic Translation Network [40], we have introduced an additional mapping function that enables the transformation from the target modality to the source modality. This integration enables the utilization of cycle consistency loss [39] as an additional regularization.

In contrast to the cycle GAN framework, we have simplified the architecture by employing a reconstructor F rather than another encoder-encoder network to achieve the mapping from the target modality to the source modality:

$$\hat{X}^S = F(\hat{X}^T) \quad (12)$$

where the F is also implemented by transformer layers.

This methodology can also be interpreted as a form of back-translation, a widely employed technique in language translation tasks [41-44].

iii. Learning objective

The cyclic transformer is trained on the paired multimodal data and labels.

The optimization objective includes two parts: reconstruction loss between two modalities and the final classification loss.

The first loss is the forward translation (mapping) loss L_f which is defined as

$$L_f = E(l_{XT}(G(X^S), X^T)) \quad (13)$$

And the second loss is the cycle consistency loss which is defined as

$$L_b = E(l_{XS}(F(G(X^T)), X^S)) \quad (14)$$

Then, the classification loss is defined as

$$L_c = E(l_c(D(G_e(X^S)), Y)) \quad (15)$$

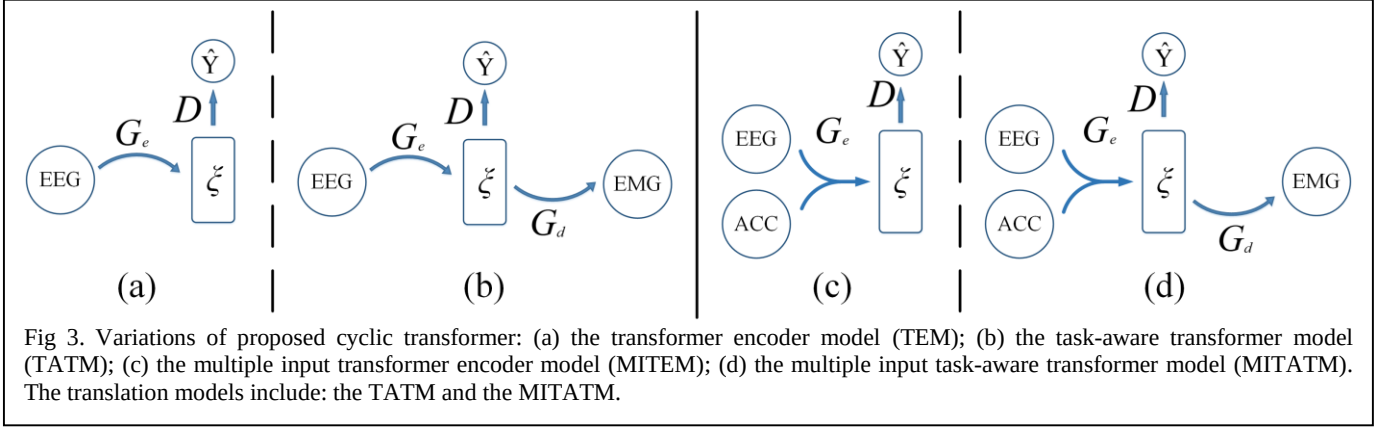
where l_{XT} , l_{XS} , were mean square error; and l_c was binary cross entropy in this paper.

The full objective is then:

$$L = L_f + L_b + L_c \quad (16)$$

The proposed cyclic transformer is trained end-to-end. Its parameters θ are learned by minimizing the full objective function

$$\theta = \arg \min_{\theta} \{L_f + L_b + L_c\} \quad (17)$$



Giving a bi-modal training set $X = \{\{X_1^S, X_1^T\}, \dots, \{X_L^S, X_L^T\}\}$, where L denotes the number of training samples, the objective of cyclic transformer is to identify the label $Y \in \{0, 1\}$ of each samples.

B. Multimodal Input Cyclic Transformer

The cyclic transformer can be extended to a multiple input version that learns an intermediate representation for multimodal input. This extension takes two modalities as inputs; one modality serves as the main modality, while the other serves as a supporting modality that aids the main modality by extracting informative features selectively.

As shown in Figure 2, the multimodal input cyclic transformer (MICT) takes two modalities as inputs. Each modality will be processed by a transformer block to have a better representation. Then, they will be fed to the cross-attention units in which the supporting modality will help the main modality extract more informative features.

The learned intermediate representation is given by the cross-attention layers and will be projected into the target modality by the decoder. Same to the cyclic transformer, a reconstructor was also added for back translation. Notably, there is only one reconstructor to reconstruct the main modality rather than multiple reconstructors. The end-task classifier is also inserted for regularization.

IV. EXPERIMENT

In this section, we introduce the multimodal dataset that was used to evaluate the TCS detection performance. We provide details about the dataset, including the collected modalities, length of recordings, and the ground-truth annotations. Moreover, we present several variations of the proposed cyclic transformer model, highlighting different architectural components. These variations were developed to showcase the effectiveness of each component. Additionally, we introduce a few benchmark models for further comparison.

A. Dataset

i. Data Collection

The data was collected in SeizeIT2, an EIT Health sponsored project. The SeizeIT2 project was approved by the ethical committee of UZ/KU Leuven (internal number:

S63631). Written informed consent was obtained for every participant.

During the project, participants were instructed to wear two Sensor Dot (SD) devices [45] to simultaneously capture multiple wearable signals. The EEG SD was attached to the upper back to record behind-the-ear EEG (BTE-EEG) and ACC data, while the EMG SD was attached to the left chest to record electrocardiography (ECG), ACC, and EMG signals from the arm. The BTE-EEG, EMG, and ECG are recorded at a sampling rate of 250 Hz while ACC is sampled at 25 Hz.

A total of 43 participants had their TCSs recorded by the SDs. Among these participants, 27 had their signals successfully recorded by both SDs, while the remaining participants only had their signals successfully recorded by the EEG SD.

To organize the data, we created two datasets: dataset A which includes 3233 hours of recording from participants who had all the signals recorded, and dataset B which comprises 1284 hours of recording from participants who had only their EEG and ACC signals recorded.

ii. Annotations

The ground truth was annotated using the clinical video-EEG, which was recorded at the same time as SD recordings. Neurologists, who were blinded to the wearable data and the output of the seizure detection algorithms, annotated all TCSs. The seizure annotations included the start and stop of the seizures, the start of the tonic phase, the start of clonic phase, and the type of the seizures (i.e., focal or generalized onset).

According to the annotations, 43 focal to bilateral TCSs (FBTCS) and 1 generalized TCS (GTCS) were included in dataset A. The average duration of TCSs in dataset A was 124 seconds (range: 55 - 661 s). 19 FBTCS were included in dataset B. The average duration of TCSs in dataset B was 183 seconds (range: 44 - 570 s).

As the focal phase is not shown in the EMG signals, only the tonic phase and the clonic phase were used to train the neural networks.

B. Benchmarks

We introduce three benchmark models for a more comparative evaluation on the proposed cyclic transformer.

The first three models are feature-based and have been

selected due to their state-of-the-art performance [17]. The first benchmark model, which is called Feat, is a feature-based unimodal model that uses EEG signals exclusively for TCSs detection. Furthermore, two multimodal models that employ late fusion techniques on feature-based unimodal models are also incorporated for comparative analysis. The first one combines EEG and EMG signals, while the other leverages the combination of EEG and ACC signals. All the feature-based unimodal models follow a consistent framework. Firstly, the incoming signals undergo preprocessing to ensure their quality and are then divided into multiple segments for further process. Subsequently, these preprocessed signal segments are subjected to feature extraction, which aims to capture relevant patterns and characteristics. Finally, the models utilize a support vector machine to classify these segments into either seizure or non-seizure categories. The information on the features used for each modality and the specific learning process of the support vector machine can be found in our previous research paper [17].

As deep learning-based approaches become increasingly prevalent in the field of seizure detection, we reproduced two deep learning-based seizure detection methods for a more comprehensive comparison. Those methods are not specifically developed for TCS detection, but serve as a general benchmark.

The first approach is ChronoNet a well-known neural network designed for particular application on EEG data in the context of seizure detection [46]. The pipeline of ChronoNet can be summarized as follows: ChronoNet first segments the input EEG data into manageable temporal windows (in our case, 4 seconds). Then one-dimensional convolutional layers are applied to the temporal windows of the EEG data, allowing the network to learn local patterns and features. Temporal patterns, such as spikes or oscillations, at different scales will be captured in this process. The resulting feature maps are flattened to create a vector representation, which is then processed through Gated Recurrent Unit (GRU) layers. The GRU captures temporal dependencies and relationships between sequential data points, allowing the model to consider the context of previous observations when making predictions. Finally, the features are passed through a predictive layer, typically a fully connected network, to generate predictions for each EEG segments window.

The second deep learning-based benchmark, known as ResNet-LSTM, demonstrated state-of-the-art seizure detection performance in a comparative review paper [47]. This model initiates by segmenting EEG signals into 4-second windows. Subsequently, it employs two-dimensional convolutions to extract features from the EEG signals. Notably, ResNet blocks [48] are incorporated in this convolutional process, aiding the model in discerning both simple and complex patterns in seizures through a flexible gradient flow enabled by skip connections. The extracted features are then passed into a Long Short-Term Memory (LSTM) network, allowing the model to capture temporal feature representations effectively.

Finally, a linear fully connected network is employed to generate predictions for each EEG segment.

Furthermore, for a more comprehensive assessment of the capability to handle missing modalities, we replicated a meta-learning framework proposed by Ma et al. [49]. This framework, inspired by a modified Model-Agnostic Meta-Learning (MAML) approach, integrates the main network, the feature reconstruction network, and the regularization network. The feature reconstruction network is introduced to approximate missing modalities by predicting weights of modality priors instead of directly generating them. These priors are learned from the modality-complete dataset using K-means clustering. The entire set of networks is jointly optimized within a Bayesian Meta-Learning framework. Consequently, the Meta-Learning framework can be trained with both EEG and EMG data but only utilizes EEG as input during deployment.

C. Ablation study

We present several variations of the proposed architecture to highlight the effectiveness of each component within the proposed cyclic transformer.

The first variation is a unimodal architecture that does not involve any translation between modalities. We call this model transformer encoder model (TEM). It consists of an encoder and a classifier, both adopting the same architecture as the ones used in the cyclic transformer. The encoder projects the input into an embedding, and the classifier generates predictions based on this embedding. This variation can be seen as a simple classification model that was trained with EEG signals as there is no translation between modalities employed.

The second variation incorporates an encoder-decoder architecture to enable translation between modalities, along with a classifier for making predictions. Similar to the previous variation, the encoder, decoder, and classifier in this variation employ the same architecture as the ones used in the cyclic transformer. We call this model task-aware transformer model (TATM). This variation is a co-learning model that learns an intermediate representation of EEG and EMG by translating between them. This variation is employed to demonstrate the advantages of back-translation as it can be viewed as a cyclic transformer without back-translation.

We also extend these two variations to include multimodal input by adapting the multimodal input cyclic transformer, i.e., the multiple input transformer encoder model (MITEM) and the multiple input task-aware transformer model (MITATM). This results in the multimodal input versions of the aforementioned variations, i.e., a simple classification model trained with EEG and ACC as well as translation model that trained with EEG, ACC, and EMG. Figure 3 shows the architecture of all the variations for comprehensive understanding and reference.

D. Implementation

Table 1. Performance evaluation of the cyclic transformer (CT), the multimodal input cyclic transformer (MICT), and the state-of-the-art benchmarks in tonic-clonic seizure detection

Model	Test Input	Metrics							
		Dataset A				Dataset B			
		Sens	FPR/24h	PPV	F1-score	Sens	FPR/24h	PPV	F1-score
Feat-EEG	EEG	100.0%	10.3	3.1%	6.0%	100.0%	116.1	0.3%	0.6%
ChronoNet	EEG	65.9%	22.4	1.0%	1.9%	64.8%	27.0	0.9%	1.8%
ResNet-LSTM	EEG	100.0%	1.3	19.9%	32.2%	73.7%	1.25	17.3%	28.0%
Meta-Learning	EEG	95.5%	4.8	6.1%	11.5%	57.9%	3.4	5.7%	10.3%
CT	EEG	100.0%	0.8	28.4%	44.2%	78.9%	0.7	28.3%	41.7%
Feat-Fusion	EEG, ACC	95.5%	1.1	20.0%	35.7%	36.8%	2.1	5.9%	10.1%
Feat-Fusion	EEG, EMG	97.7%	0.4	43.0%	59.7%	N/A	N/A	N/A	N/A
MICT	EEG, ACC	100.0%	0.4	43.6%	60.9%	84.2%	0.6	34.8%	49.2%

Abbreviations: EEG, electroencephalography; EMG, electromyography; ACC, accelerometer; Sens, sensitivity; FPR/24h, false positive rate per 24 hours; PPV, positive predictive value (precision).

Kindly note that only the Feat-fusion of EEG and EMG necessitates the use of two sensors during deployment, making it not applicable to Dataset B.

The signals underwent preprocessing before being fed into the proposed cyclic transformer and its variations. Initially, the signals were cut into 60-second segments and then transformed into the time-frequency domain via stockwell transform. The frequency range is from 3 Hz to 64 Hz for EEG and EMG, and 0 Hz to 12 Hz for ACC. Each time-frequency matrix was then divided into 120 blocks in time axis to decrease the computational complex. As a result, a time-frequency matrix is $X \in R^{d \times f}$, where d denotes the time indices and f denotes the frequency bins, was obtained and was used as the input of the neural network.

To enhance the training dataset with more seizure samples, a 59-second overlap was introduced between two consecutive segments of the signal. Given the abundance of non-seizure samples, random under sampling was applied to create a balanced training set. For a more efficient prediction process, signal segments were adjusted to have no overlaps between consecutive segments during deployment.

We implemented the proposed cyclic transformer and its variations in PyTorch [50]. The cyclic transformer uses 4 transformer layers for the encoder, 2 transformer layers for the decoder, and 2 transformer layers for the reconstructor. In the transformer layers, the number of multi heads was set to 4 and hidden units in the feedforward network was set to 1024.

The cyclic transformer and its variations are trained by an Adam Optimizer [51] with a learning rate of 5×10^{-5} and a l2 regulation rate of 1×10^{-4} . The neural networks were trained 25

epochs with the data represent in a batch size of 32.

For each seizure detection approach, a leave-one-participant-out cross-validation approach is conducted. In each fold of the cross-validation, the data from one participant is set aside as the test set to evaluate the trained model's performance. Subsequently, the data from the remaining participants was divided into a training set and a validation set. Notably, the data from each participant exclusively serves as either the training set or the validation set, ensuring that the data from the same participant is present in only one of these datasets. The training set and validation set are then utilized for model training and evaluating the model fit during the training process, respectively.

The early stopping technique [52] was applied by selecting the model with the minimum loss on the validation set during the training process.

E. Evaluation and Metrics

The performance of each model was evaluated by conducting a leave-one-participant-out cross-validation on dataset A. In each fold of the cross-validation, the data from one participant is set aside as the test set to evaluate the trained model's performance; the data from the remaining participants was then used for training and validation. Subsequently, an independent test set, namely dataset B, was employed to further evaluate the models. Notably, dataset B exclusively consists of EEG and ACC recordings, making it inapplicable for training purposes.

Table 2. Performance evaluation of the cyclic transformer and its variations in tonic-clonic seizure detection

		Metrics							
		Dataset A				Dataset B			
Model	Test Input	Sens	FPR/24h	PPV	F1-score	Sens	FPR/24h	PPV	F1-score
TEM	EEG	100.0%	1.2	22.0%	36.1%	78.9%	0.7	27.3%	40.5%
TATM	EEG	97.7%	0.9	25.9%	41.0%	78.9%	1.0	22.4%	34.9%
CT	EEG	100.0%	0.8	28.4%	44.2%	78.9%	0.7	28.3%	41.7%
MITEM	EEG, ACC	100.0%	1.0	24.9%	39.8%	78.9%	1.0	22.7%	35.3%
MITATM	EEG, ACC	97.7%	0.7	30.3%	46.2%	78.9%	0.7	29.4%	42.9%
MICT	EEG, ACC	100.0%	0.4	43.6%	60.9%	84.2%	0.6	34.8%	49.2%

Abbreviations: EEG, electroencephalography; EMG, electromyography; ACC, accelerometer; Sens, sensitivity; FPR/24h, false positive rate per 24 hours; PPV, positive predictive value (precision).

The translation methods include the TEM, CT, MITEM, and MICT.

There are two scoring methods commonly employed for evaluating time series pattern recognition tasks. The first, widely used in machine learning tasks, is sample-based scoring. In this method each signal segment is treated as an individual sample, and metrics are calculated based on whether a sample is correctly classified.

However, misleading conclusion can be made by utilizing sample-based scoring in time series pattern recognition tasks. Notably, it tends to assign more weight to events with longer durations (i.e., containing more samples/signal segments), as each segment is evaluated independently. This approach may lead to the overlooking of shorter events [53].

In tasks such as seizure detection, temporal localization is critical, emphasizing the alignment of event start and stop times between hypothesis annotations and ground truth annotations. Therefore, term-based scoring metrics are widely utilized in this context [6, 7, 9, 10, 12]. These metrics consider the overlap between ground-truth annotations and algorithm-generated hypothesis annotations, counting each event only once during the evaluation. The true positives (TP), false negative (FN), and false positive (FP) were then defined as

- TP represents the detected TCS. It signifies that a seizure was detected when there were positive predictions within the duration of ground-truth annotation.
- FN is an undetected TCS.
- FP represents a continuous sequence of positive predictions that has no overlap with the ground-truth annotation. In other words, it refers to instances where the algorithm incorrectly identifies a seizure when there is no corresponding TCS in the ground-truth annotation

Studies in seizure detection domain typically report performance in terms of sensitivity and FP rate to provide intuitively reflection on task performance. To gain a more

comprehensive understanding, precision and F1-score are also employed. Based on the aforementioned definitions, sensitivity, FP rate, precision, and F1-score were calculated according to the definition in [54] to evaluate the performance of each model.

V. RESULTS AND DISCUSSION

A. Overview

Table 1 illustrates performance of the cyclic transformer, MICT, and the state-of-the-art benchmarks in TCS detection.

Notably, the cyclic transformer outperforms other approaches, particularly when the test input is unimodal EEG, highlighting the advantages of incorporating multiple modalities during training. Even with an equal number of modalities for training, Meta-Learning falls behind the cyclic transformer, indicating challenges in reconstructing the missing modality. This difficulty in effective reconstruction is attributed to the high variability of EEG and EMG. The cyclic transformer addresses this issue by circumventing the reconstruction of the missing modality during deployment.

Among the three approaches which utilize only EEG for both training and testing, ResNet-LSTM stands out, demonstrating superior performance. This highlights its capability to extract and process more complex features from EEG signals.

Among the approaches which use multiple modalities for training and testing, the MICT achieved the highest performance on both dataset A and dataset B. Compared to the feature-based EEG & ACC model which use the same amount of input modalities at test, MICT consistently achieved superior results on both datasets, implying that better performance can be achieved by employing more modalities for training.

Table 3. Effect of varying source and target modalities on seizure detection performance in the cyclic transformer

Model	Test input	Metrics			
		Sens	FPR/24h	PPV	F1-score
TEM	EEG	100.0%	1.2	22.0%	36.1%
TATM	EEG	97.7%	0.9	25.9%	41.0%
CT	EEG	100.0%	0.8	28.4%	44.2%
TEM	EMG	86.4%	8.9	3.1%	5.9%
TATM	EMG	86.4%	7.0	3.9%	7.4%
CT	EMG	81.8%	6.6	3.9%	7.4%

Abbreviations: EEG, electroencephalography; EMG, electromyography; Sens, sensitivity; FPR/24h, false positive rate per 24 hours; PPV, positive predictive value (precision).

The results are derived from dataset A, as dataset B lacks EMG data and is thus not applicable for this particular analysis.

The feature-based EEG & EMG model achieved the second best performance on dataset A, demonstrating a superior performance compared to the other models. However, it is important to note that the feature-based EEG & EMG model relies on two sensors and consequently was not applicable on dataset B. This highlights a limitation of such simple fusion models, as they depend on multiple sensors and have no robustness to the loss of modalities.

B. Advantages of Cyclic Translation: An Ablation Study

Table 2 presents the outcomes of the ablation study, providing insights into the functionality of the cyclic transformer.

i. Advantages of Learning with Multiple Modalities

The results of the ablation study, as presented in Table 2, demonstrate the impact of incorporating multiple modalities as input. Notably, all the models successfully operated using only a single sensor, while showing varied performance outcomes. These models can be categorized into three distinct groups, based on the modalities they utilized for testing:

As shown in Table 2, TEM and MITEM fall respectively at the bottom of their group in term of TCS detection performance. This is due to the fact that they are trained with less modalities compared to the other models in their group. The lack of translation techniques hinders the ability of using multiple modalities during the training phase, leading to inferior performance compared to translation models.

The superior performance of translation models can be attributed to their capacity to incorporate more modalities during training, leading to the acquisition of more discriminative and informative representations of the inputs.

Notably, among the translation models, those incorporating the back-translation technique achieve better performance.

ii. Advantages of Back-translation

Results showed that all the translation models achieved better performance than other deep-learning models. Greater improvement was observed when back-translation is applied.

The advantages of back-translation can be intuitively explained as follows

- Back-translation serves as a regularization technique by enforcing consistency between the representations of different modalities.
- Back-translation facilitates the integration of information from multiple modalities and as such, the model can benefit from the diverse information. By cyclically translating between modalities, the model can capture complementary information and learn to combine the modalities effectively.

iii. Advantages of Multiple Input Cyclic Transformer

The results show that incorporating more modalities can notably enhance performance. This is also confirmed by the cyclic transformer. The MICT achieved better performance compared to the cyclic transformer. This shows that performance may be improved when more information is fed to the model.

The MICT slightly outperformed the state-of-the-art, which is a feature-based model that combines EEG and EMG for seizure detection. Notably, the MICT requires only one sensor during deployment, whereas the feature-based model necessitated two sensors. This crucial distinction renders the MICT applicable to dataset B, whereas the feature-based model falls short due to its sensor requirements.

These results underscore the robustness of the MICT in handling sensor malfunction and highlight its cost-effectiveness thanks to the utilization of fewer sensors during deployment. The ability of the MICT to adapt to data with missing modalities reinforces its potential as a practical and efficient solution in real-world seizure detection scenarios.

Table 4. Model Complexity and Average Detection Delays of Each Deep Learning Model

	Duration of Test Input	Average Detection Delay	N of Parameters	Execution Time
ChronoNet	4 seconds	39 seconds	1.2 million	0.001 seconds
ResNet-LSTM	4 seconds	74 seconds	3.9 million	0.001 seconds
Meta-Learning	60 seconds	93 seconds	1.4 million	0.089 seconds
CT	60 seconds	98 seconds	0.9 million	0.090 seconds
MICT	60 seconds	97 seconds	1.5 million	0.101 seconds

Abbreviations: N of Parameters, number of parameters; CT, cyclic transformer.

C. Select the Optimal Input Modality

The impact of varying the source and target modalities was also investigated, as presented in Table 3. The most favorable results were achieved when EEG served as the source modality and EMG as the target modality.

Unlike the models that treat all input modalities equally, the cyclic transformer performed differently when source and target modalities are interchanged. This aligns with the nature of the translation architecture.

In this context, the modality employed solely during training is leveraged to provide support to the other modality through the transfer of knowledge. The knowledge transfer between modalities leads to distinct performance outcomes depending on the employed source and target modalities.

The observed performance variations based on modality variations contribute to a deeper understanding of the translation architecture's dynamics and underscore the importance of considering the selection of modalities in achieving optimal performance.

D. Model Complexity and Detection Delay

Given the imperative for seizure detection algorithms to be lightweight and applicable to wearable devices, we assess the model complexity of each seizure detection approach by calculating the amount of trainable parameters and testing the needed execution times during deployment. We, therefore, conducted inference experiments on an Intel Core i7 CPU.

In general, all the seizure detection approaches are able to process the input within 0.1 seconds, showcasing their capacity for real-time applications.

Table 4 illustrates that the proposed cyclic transformer has the smallest amount of parameters among all the deep learning approaches. However, the input requires transformation into the time-frequency domain, resulting in a longer average processing time. Despite this extended preprocessing time, it still achieves an average processing time of 0.0904 seconds per sample, making it suitable for real-time applications.

The MICT, which incorporates additional transformer layers, exhibits a moderate level of complexity with approximately 1.5 million parameters. Like the cyclic transformer, time-frequency transformation is also necessary, extending the average processing time to 0.1009 seconds.

Recognizing the importance of a lightweight model for wearable applications, we explored the possibility of further streamlining the model by reducing the number of transformer layers for encoding, decoding, and reconstruction.

Experimental results indicate a reduction in parameters to 0.8 million when fewer layers are used in the encoder, decoder, and reconstructor of MICT., but this reduction impacts performance. While its sensitivity remains at 100.0%, the false positive rate (FPR) increases to 0.5 per 24 hours, resulting in a decrease in the F1-score from 60.9% to 55.0%. Similarly, by streamlining its architecture, the cyclic transformer reduces its parameters from 0.9 million to 0.5 million. However, sensitivity decreases from 100.0% to 93.2%. The FPR also decreases from 0.8 to 0.6, maintaining the F1-score at 48.5%. This observation aligns with other studies, emphasizing the crucial role of appropriate complexity in ensuring good performance.

As shown in Table 4, the majority of the models, except for ChronoNet, which demonstrates underwhelming detection performance, exhibit a prolonged detection delay. This indicates the inherent trade-off between ensuring short detection delay and achieving accurate detection. However, we would like to highlight that cardiopulmonary resuscitation (CPR) are typically initiated within 1 to 3 minutes after the end of a TCS to avoid sudden unexpected death in epilepsy [1]. Gutierrez et al. suggested that an alert should be raised approximately 80-120 seconds from the onset of TCSs [55]. Detecting a seizure 97 seconds after its onset aligns with this critical timeframe for timely intervention. Accurate detection takes precedence over achieving an extremely short delay in this scenario, as missing the detection of a TCS can result in severe consequences.

VI. CONCLUSION

In this paper, we introduced the cyclic transformer, a novel approach for learning robust representations in TCSs. The cyclic transformer operates by translating from source modalities to target modalities, allowing it to learn an intermediate representation without relying on the presence of target modalities during testing. This unique characteristic endows the cyclic transformer with robustness against missing

target modalities. Additionally, we incorporated the back-translation technique as a regularization method within the cyclic transformer, which further improved the discriminative of the learned representation.

Experimental results demonstrated that the cyclic transformer, empowered by back-translation, achieved competitive results in TCS detection compared to state-of-the-art methods while requiring fewer sensors to record the input signals. This dual advantage of robustness and reduced sensor requirements makes the cyclic transformer a promising and cost-effective solution for TCS detection.

In conclusion, our proposed cyclic transformer not only demonstrates robustness in handling missing modalities but also offers an effective approach for accurate TCS detection. The performance highlights its value as a tool in long-term monitoring of epilepsy patients with TCSs.

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